YASH ARYAN

Bhopal, Madhya Pradesh, India | +91 9125200020 | yasharyanchunar90@gmail.com
LinkedIn: https:/linkedin.com/in/yasharyan90 | GitHub: https:/github.com/yasharyan90 | LeetCode: https:/leetcode.com/u/YashAryan90 | Portfolio Web Link - https://my-portfolio-web-murex.vercel.app/

PROFESSIONAL SUMMARY

B.Tech CSE (Cloud Computing and Automation) undergraduate (CGPA 8.0/10) targeting backend and cloud engineering roles. Strong in C++, DSA (500+ problems solved), OS, networking, and Linux, with hands-on experience building real-time, scalable systems using Go, Node.js, WebSockets, Redis, PostgreSQL, Docker, and AWS. Built production-grade projects involving concurrency, authentication, and system debugging.

EDUCATION

Vellore Institute of Technology (VIT Bhopal)
Expected Graduation: Sept 2028 B.Tech in CSE(CLOUD COMPUTING AUTOMATION) **CGPA: 8.0 / 10.0**
Relevant Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Object-Oriented Programming, Cloud Computing Tools & Services, AI Agents, Unix Tools and Scripting

TECHNICAL SKILLS

Programming Languages:C++, Python, Go, JavaScript, TypeScript, SQL
Core Computer Science: Data Structures & Algorithms, Oops, OS, CN, System Design, Debugging
Systems Linux: Linux/Unix, Shell Scripting, Unix Tools, Process Management, Networking Fundamentals, Git
Backend Systems Development: Node.js, Express.js, FastAPI, REST APIs
Databases: PostgreSQL, MySQL, MongoDB, Redis, DynamoDB
Cloud DevOps: AWS (EC2, S3, DynamoDB), Docker, Docker Compose, CI/CD
Frontend: React.js, Vite, TypeScript, Tailwind CSS, PWA
Soft Skills: Problem-Solving , Team Collaboration , Clear Communication(Hindi & English)

PROJECT & RELEVENT TEAM EXPERIENCE

VIT Live — Real-Time Campus Engagement Platform Personal Project | GitHub
Go (Fiber), React 19, TypeScript, PostgreSQL, Redis Pub/Sub, WebSockets, Razorpay, Tailwind CSS, Framer Motion, PWA

- Architected a full-stack real-time platform with 5-role RBAC, live announcements, emergency alerts, polls, and club feeds using WebSockets and Redis Pub/Sub, achieving sub-second message delivery.
- Integrated Razorpay payments with HMAC-SHA256 signature verification, QR-code event ticketing, and live attendee check-in.
- Secured the platform with JWT authentication, rotating refresh tokens, OTP verification, anonymous polling, audit logging, and CI-tested REST APIs.
- Shipped an installable React PWA with offline support, Firebase Cloud Messaging push notifications, admin dashboard, scheduled publishing, and S3-compatible object storage across a 13K+ LOC monorepo.

Unified Health Intelligence Platform (UHIP) — Cloud-Native Healthcare Platform Team Project

- Designed and deployed a multi-service software architecture integrating React, Express.js, FastAPI, PostgreSQL, and MongoDB services using Docker and Docker Compose.
- Configured container networking, persistent storage, service dependencies, environment management, and inter-service communication for a reproducible Linux-based deployment environment.
- Performed system-level debugging, integration testing, failure analysis, and deployment troubleshooting to improve reliability across multiple services.
- Led containerization and deployment activities for the team project; awarded an S Grade and presented the system at the College Project Exhibition.

ColCode- Real-Time Collaborative Code Editor Personal Project | Production Docker Deployment
React, TypeScript, Monaco Editor, Yjs (CRDT), WebSockets, xterm.js, Node.js, Fastify, PostgreSQL, Redis, Docker, Caddy

- Built a real-time collaborative code editor using Yjs CRDT and WebSockets, enabling conflict-free concurrent editing with live cursors, selections, user presence, and zero lost keystrokes during simultaneous typing.
- Engineered a secure code-execution sandbox where each run launches a single-use Docker container with network isolation, read-only root filesystem, dropped capabilities, and cgroup CPU, memory, PID, and wall-clock limits; supports Python, JavaScript, TypeScript, Ruby, and Go with streamed stdout/stderr.
- Enforced project-level RBAC (owner/editor/viewer) using GitHub OAuth and JWT sessions, with authorization checked during WebSocket upgrades and CRDT snapshots persisted to PostgreSQL.
- Load-tested the synchronization server at 3,500 ops/s across 100 rooms and 500 concurrent clients, achieving 100shipped 150 automated checks covering synchronization, access control, sandbox limits, browser E2E testing, and production Docker Compose deployment behind Caddy with HTTPS.

CERTIFICATIONS & ACHIEVEMENTS

- **Applied Machine Learning in Python** — University of Michigan (Coursera 88.88%): Predictive Modeling, Scikit-Learn.
- **Cloud Computing and Distributed Systems** — NPTEL, IIT Kanpur (Elite + Silver, 78%): Virtualization, Scalability.
- **Introduction to Internet of Things** — NPTEL, IIT Kharagpur (Elite, 77%): IoT Architecture, Embedded Systems.
- **Big data Computing** — NPTEL, IIT Kanpur (Elite 70%): Hadoop , Map Reduce , Apache Spark
- **Solved 500+ Data Structures & Algorithms problems** on LeetCode and GeeksforGeeks; following Striver's A2Z DSA Sheet.